IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies:

- Data Collection & Wrangling: Extracted SpaceX launch data using REST APIs (requests library) and web scraped Wikipedia tables (BeautifulSoup); handled missing values and created a binary target label Class (1 for success, 0 for failure).
- Exploratory Data Analysis (EDA): Performed initial trends analysis using Matplotlib/Seaborn visualizations and executed SQL queries to aggregate key performance metrics.
- Geospatial & Interactive Analytics: Built Folium maps to analyze launch site proximity and safety zones; created a Plotly Dash interactive dashboard to isolate payload ranges and site performance.
- Predictive Modeling: Scaled features using StandardScaler and trained four classification algorithms (Logistic Regression, SVM, Decision Tree, KNN) tuned via 10-fold cross-validation GridSearchCV.

Summary of All Results:

- Historical Trends: Launch success rates improved dramatically over time, growing from 0% (2010–2013) to over 85% by 2020. Kennedy Space Center (KSC LC 39A) achieved the highest overall site success rate at 77.27%.
- Model Performance: Logistic Regression, SVM, and KNN tied for top performance on the unseen test dataset, each achieving an 83.33% accuracy score.
- Error Profile: Confusion matrix evaluation showed zero false negatives and 3 false positives (predicting a successful landing for a flight that failed).
- Business Value: Predictive capability enables accurate pricing models ($62M for reused Falcon 9 launches vs. $165M+ for traditional expendable competitors).

Introduction

Project Background and Context:

- SpaceX has disrupted the commercial aerospace industry by standardizing the recovery and reuse of Falcon 9 first-stage rocket boosters.
- Commercial rocket launches traditionally cost over $165 million due to single-use components, whereas SpaceX advertises launches at around $62 million by recovering first-stage cores.
- Predicting first-stage landing outcomes is vital for estimating reuse probability, evaluating mission risk, and providing competitive commercial pricing proposals.

Problems to Answer:

- Predictive Capability: Can machine learning classification models accurately predict if a Falcon 9 first stage will successfully land on a ground pad or drone ship?
- Feature Impact: How do flight factors—such as Payload Mass, target Orbit type (e.g., GTO, LEO, VLEO), and Launch Site location—correlate with booster landing success rates?
- Geospatial Constraints: What proximity factors (distance to coastlines, railways, highways, and urban areas) drive launch site operational selection?
- Model Optimization: Which classification model (Logistic Regression, Support Vector Classifier, Decision Tree, or K-Nearest Neighbors) generalizes best to unseen future launch data?

Section 1

Methodology



Methodology

Executive Summary

Data collection methodology:

- SpaceX REST API: Extracted raw launch data, core info, payload specs, landpads, and launchpad details using Python's requests library across 5 distinct endpoints.
- Web Scraping: Built automated HTML table parsers using BeautifulSoup to extract historical Falcon 9 launch records and booster specifications directly from Wikipedia.

Data Wrangling & Processing:

- Cleaned missing values (such as replacing nulls in PayloadMass with the column mean) and structured JSON payloads into Pandas DataFrames.
- Filtered bad outcomes, calculated mission landing success criteria, and constructed the binary target vector Class (1 for success, 0 for failure) yielding a 66.67% overall success baseline.

Exploratory Data Analysis (EDA):

- Visualizations: Analyzed relationships between FlightNumber, PayloadMass, Orbit types, and LaunchSite using Seaborn and Matplotlib.
- SQL Queries: Executed structured queries to compute payload totals, average masses per booster version, landing outcomes, and date filtering.

Interactive Visual Analytics:

- Folium Geospatial Maps: Plotted launch site circles, marker clusters for outcomes, distance lines (PolyLine), and mouse position tracking for coastal proximity analysis.
- Plotly Dash Dashboard: Built an interactive web application featuring site dropdowns, pie charts, payload range sliders, and scatter plots.

Predictive Classification Modeling:

- One-hot encoded categorical variables into an 83-feature matrix and applied StandardScaler normalization.
- Trained and hyperparameter-tuned four classifiers (Logistic Regression, SVM, Decision Tree, and KNN) using 10-fold Cross-Validation GridSearchCV.
- Evaluated models on an unseen 18-sample test dataset using accuracy scores and confusion matrices.

Data Collection

- **Endpoints used: Which SpaceX API URLs that I’m called:** "https://api.spacexdata.com/v4/rockets/”, "https://api.spacexdata.com/v4/launchpads/", "https://api.spacexdata.com/v4/payloads/" , "https://api.spacexdata.com/v4/cores/" and "https://api.spacexdata.com/v4/launches/past" .

- **Methods and libraries: Use of `requests.get()`, handling of response status:** (`response.status_code == 200`), and conversion to JSON (`response.json()`).

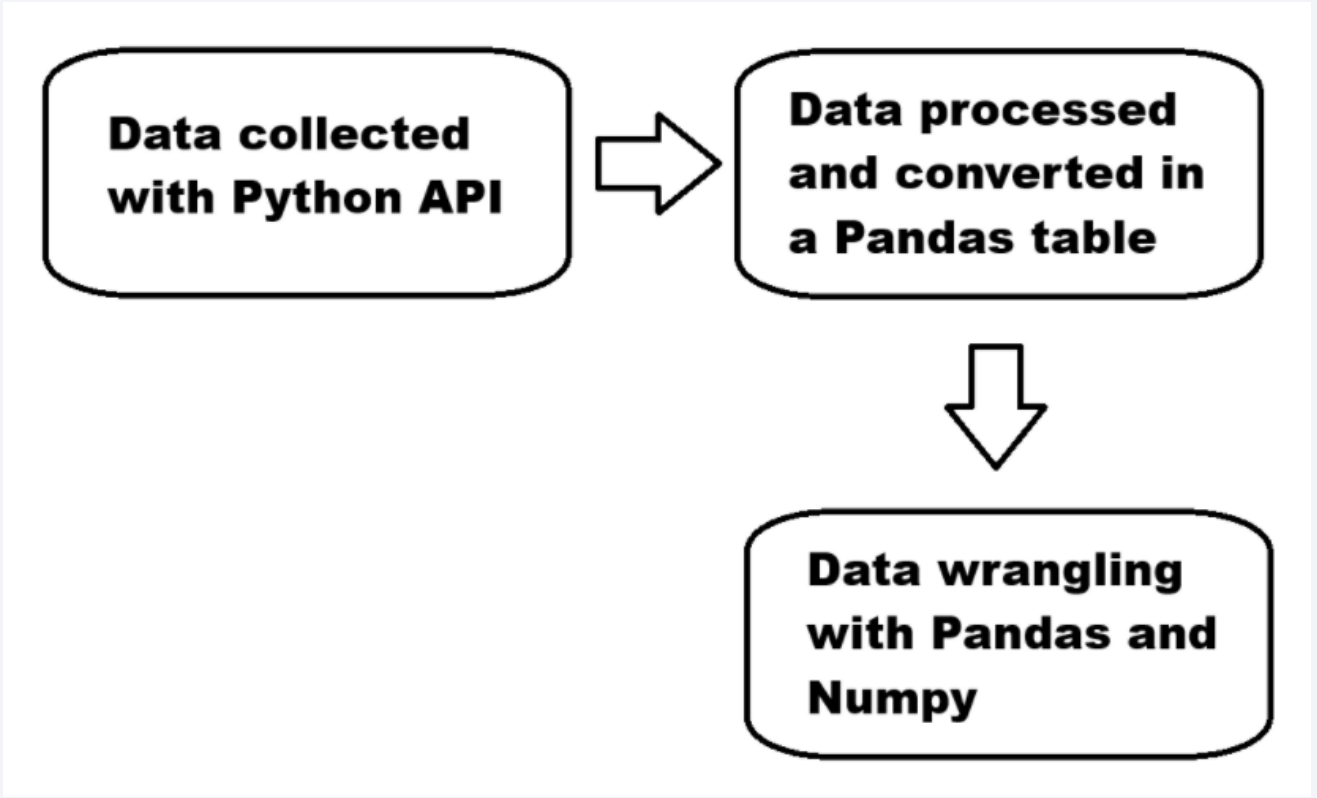
- **Parameters and filters: Whether I’m used POST requests with search payloads:** `longitude`, `latitude`, `name`,`orbit`, `flight`, `grindfins`, `reused`, `legs` and `landpad` where used as filters j the URLs during the web scrapping process.

- **Normalization/Data processing:** We calculated the mean of the values in the PauloadMass column to perform replacements for the fields with empty values.

Data Collection – SpaceX API

- The data was collected via an API using the Python 'Requests' library, as shown in the Jupyter Notebook "1_jupyter-labs-spacex-data-collection-api".
- Once collected, the data was processed from its raw HTTP format into Python Pandas tables; a function named "getCoreData" was used for this purpose.
- Subsequently, Pandas—in conjunction with the NumPy and Datetime libraries was used for further data cleaning, such as handling missing values.

GitHub URL: https://github.com/DougAugSilva/IBM_Final_Project/blob/main/1_jupyter-labs-spacex-data-collection-api.ipynb



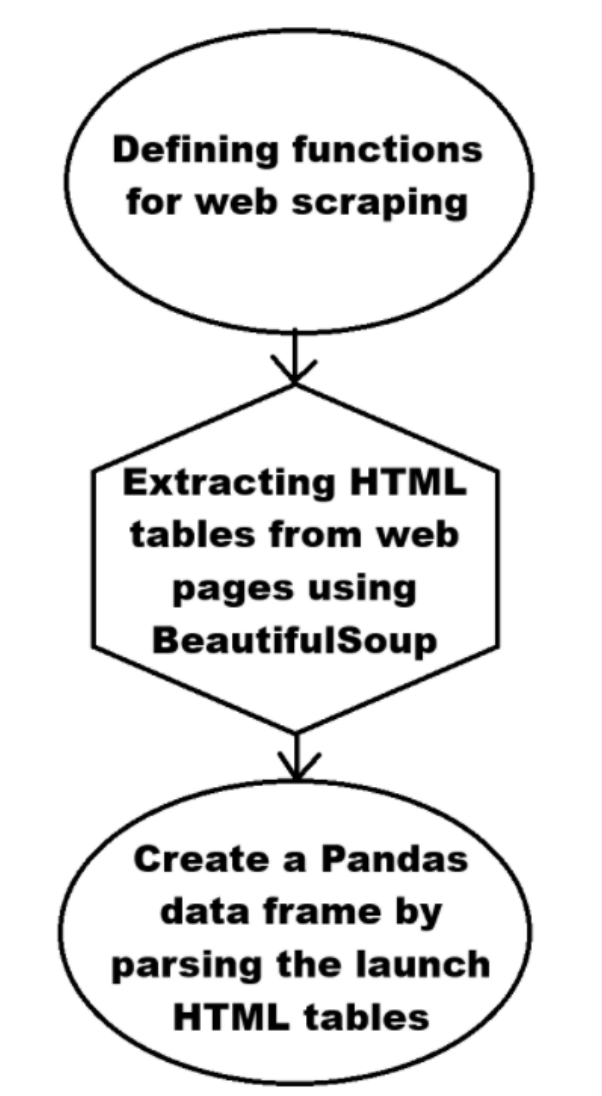
Data Collection - Scraping

The following functions have been defined to assist in extracting data via web scraping:

- **date_time:**
  - This function returns the data and time from the HTML table cell
  - Input: the element of a table data cell extracts extra row.
- **booster_version:**
  - This function returns the booster version from the HTML table cell
  - Input: the element of a table data cell extracts extra row
- **landing_status:**
  - This function returns the landing status from the HTML table cell
  - Input: the element of a table data cell extracts extra row
- **get_mass:** This function normalizes the mass by converting it to kg, in case it is in a different unit of measurement.
- **extract_column_from_header:**
  - This function returns the landing status from the HTML table cell.
  - Input: the element of a table data cell extracts extra row.

GitHub URL:

https://github.com/DougAugSilva/IBM_Final_Project/blob/main/2_jupyter-labs-webscraping.ipynb



Data Wrangling

Based on the analysis performed in this notebook, here is a concise summary of the 5 critical phases of our data wrangling pipeline:

1) **Identifying Missing Values:** Analyzed the dataset for completeness and identified that the LandingPad attribute contains approximately 28.89% missing values, while other columns are complete.

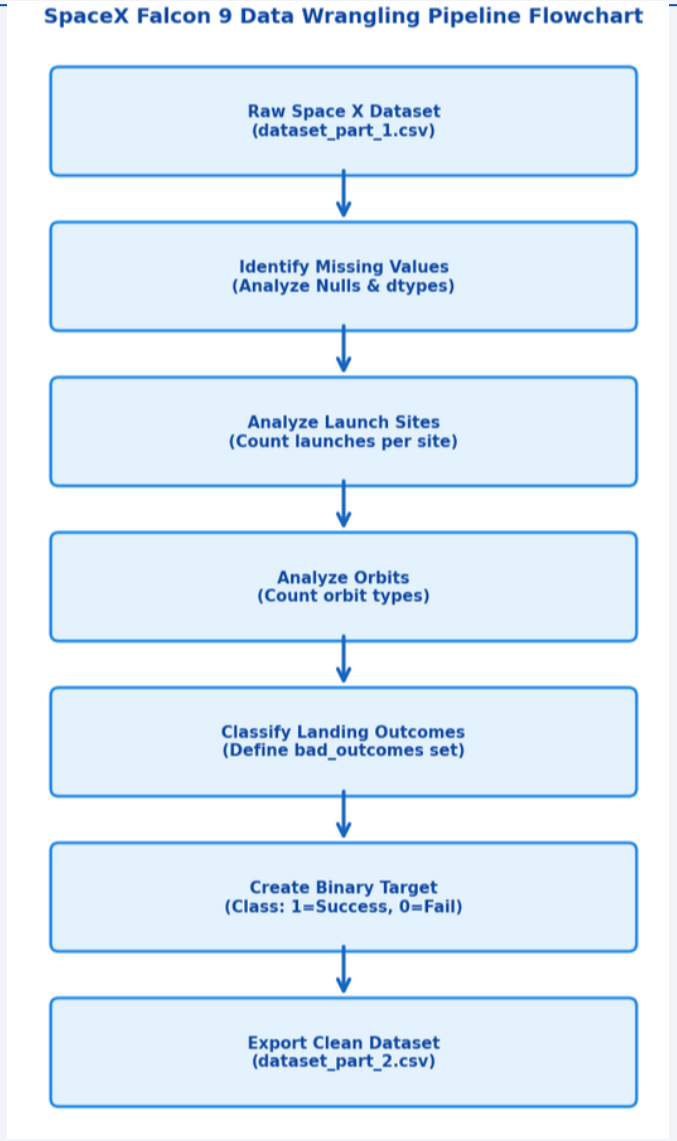
2) **Analyzing Launch Sites:** Explored launch facilities using value_counts() on the LaunchSite column, identifying the primary sites: CCAFS SLC 40 (55 launches), KSC LC 39A (22 launches), and VAFB SLC 4E (13 launches).

3) **Analyzing Orbits:** Calculated the frequency of target orbits (e.g., GTO, ISS, VLEO, PO) to understand mission distribution, noting that GTO is the most frequent transfer orbit.

4) **Classifying Landing Outcomes:** Identified and isolated landing outcomes into successful and unsuccessful categories. We defined a set of bad_outcomes representing unsuccessful landings: {'False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None'}.

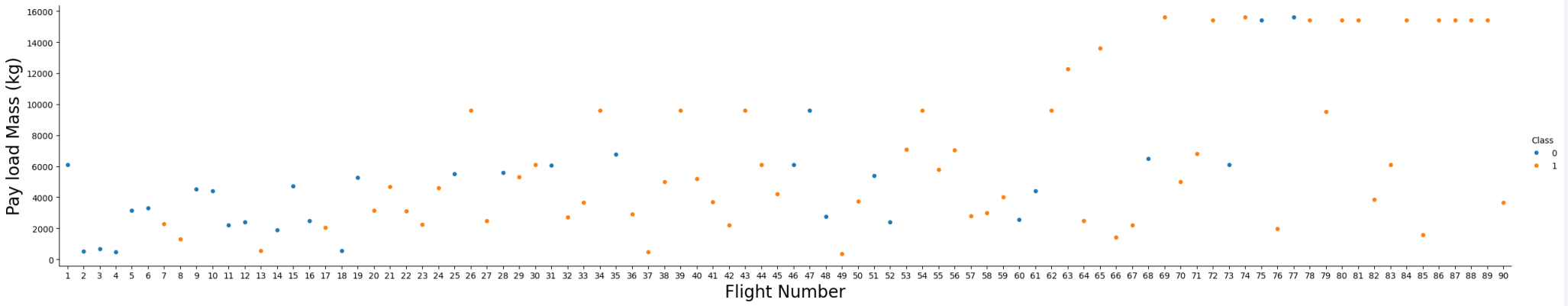
5) **Creating Binary Target & Exporting Dataset:** Formulated the binary Class target label (where 1 indicates a successful first-stage landing, and 0 indicates a failed landing) yielding a success rate of 66.67%. Finally, the prepared data is ready for export as dataset_part_2.csv.

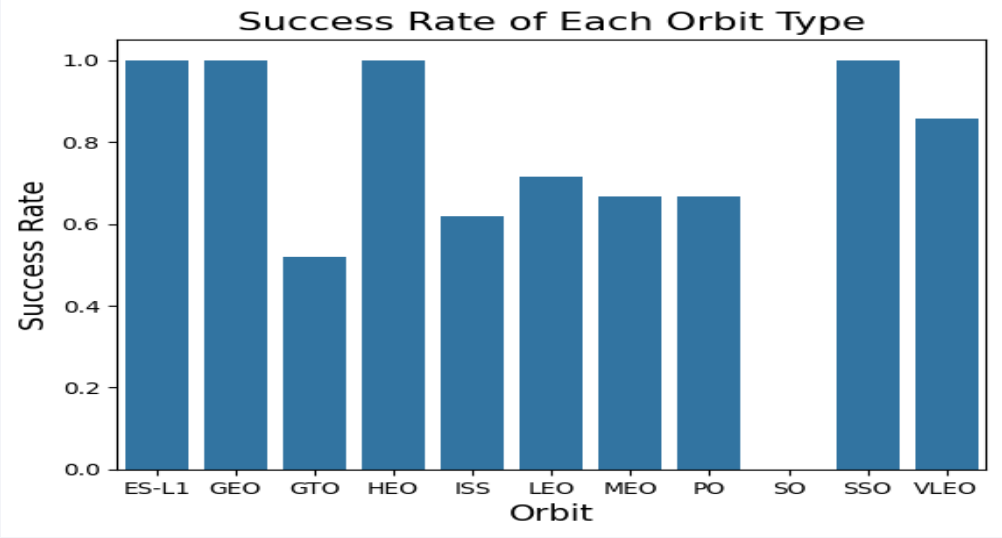
GitHub URL: https://github.com/DougAugSilva/IBM_Final_Project/blob/main/3_labs-jupyter-spacex-Data%20wrangling.ipynb

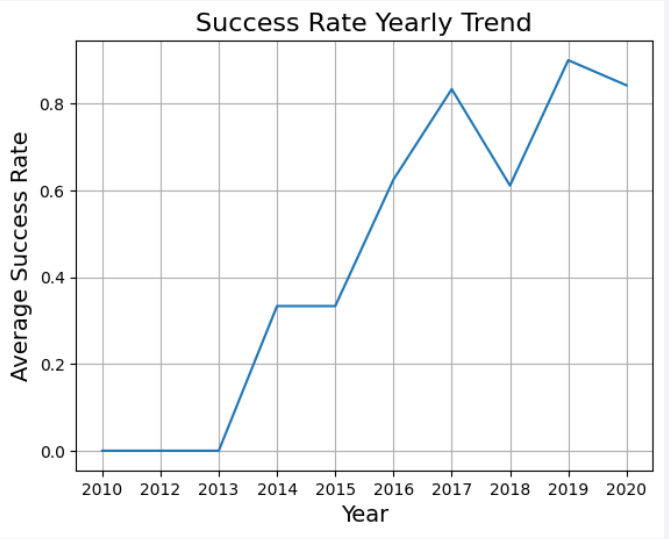


EDA with Data Visualization

We can plot out the FlightNumber vs. PayloadMassand overlay the outcome of the launch.
I also create a bar chart for the sucess rate of each orbit.
Other plot is a line chart with x axis to be Year and y axis to be average success rate, to get the average launch success trend.







EDA with SQL

- SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
- SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
- SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
- SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
- SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
- SELECT Booster_Version FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
- SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
- SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
- SELECT substr(Date, 6, 2) AS Month, "Landing_Outcome", Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)';
- SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Outcome_Count DESC;

Build an Interactive Map with Folium

- **Circles (folium.Circle):** They serve to visually highlight the immediate area of each base and provide a simple spatial scale for understanding the size of the facilities in relation to their surroundings.
- **Static Text Labels (folium.map.Marker with DivIcon):** Allows for the direct and immediate reading of each launch base's name right on the map, without the need to click on an element to open a popup, thereby improving navigability.
- **Marker Cluster (MarkerCluster):** Since there are dozens of launches for each of the same 4 launch sites, plotting every marker individually would make the map cluttered and unreadable. The clusterer groups together points that are geographically close, depending on the zoom level.
- **Custom Success and Failure Markers (folium.Marker + folium.Icon):** It allows for a quick, visual identification of the overall success rate for each launch site. By zooming in, we can see the color proportions for each individual point.
- **Linhas de Distância (folium.PolyLine):** It allows for a visual analysis of the geographical proximity of the bases and supports the investigation of hypotheses (for example, why bases are typically built very close to coastlines or far away from major urban centers).

GitHub URL: https://github.com/DougAugSilva/IBM_Final_Project/blob/main/6_lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- **Evaluate Site Performance:** The Dropdown + Pie Chart combination allows users to compare total successful launch counts across all locations or examine the precise success-to-failure ratio of any single site. This directly helps identify which site has the highest number of total successes and which has the highest success rate.

- **Analyze Payload Impact on Success Rates:** The Range Slider + Scatter Plot interaction allows users to isolate payload mass windows to pinpoint which specific ranges yield the highest or lowest mission success rates.

- **Assess Booster Version Reliability:** Color-coding the scatter plot points by Booster Version Categorymakes it easy to visually track how different Falcon 9 iterations (v1.0, v1.1, FT, B4, B5) perform across various payload weights and launch sites.

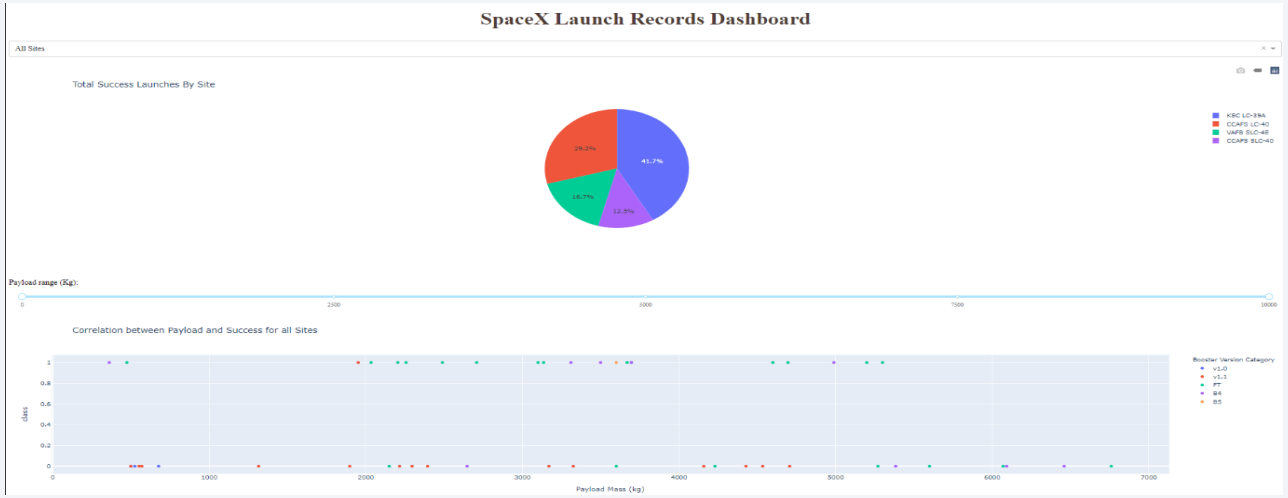
GitHub URL: https://github.com/DougAugSilva/IBM_Final_Project/blob/main/Dashboard_Instructions.md

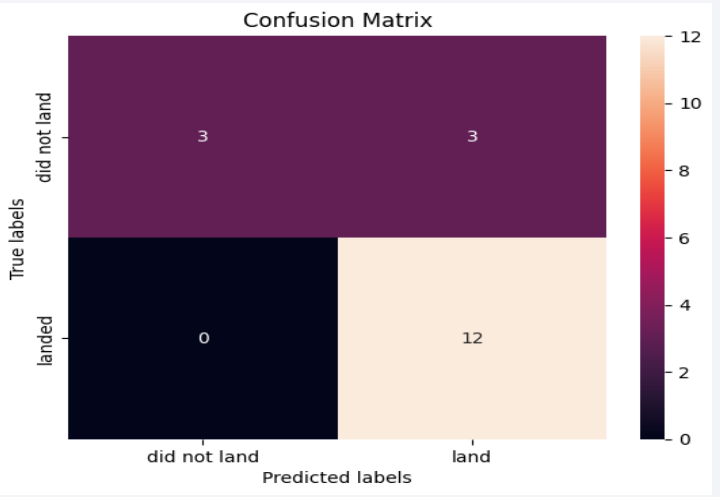
Predictive Analysis (Classification)

- **Data Ingestion & Preparation:** Loaded Falcon 9 historical landing dataset (dataset_part_2.csv) and one-hot encoded feature matrix (dataset_part_3.csv).
- **Target Vector Extraction:** Extracted the target landing outcome label (Class) and converted it into a NumPy array Y.
- **Feature Standardization:** Scaled the feature matrix X using StandardScaler to bring all features to a common scale, ensuring stable model training.
- **Train-Test Split:** Stratified the dataset into 80% training data and 20% testing data (creating a robust validation split of 18 unseen test instances).
- **GridSearchCV Optimization:** Tuned hyperparameters using a 10-fold Cross-Validation grid search to prevent overfitting on the training set.
- **Multi-Model Evaluation:** Configured and trained four distinct classifiers: Logistic Regression, Support Vector Classifier (SVC), Decision Tree, and K-Nearest Neighbors (KNN).
- **Final Selection:** Evaluated unseen test dataset performance and confusion matrices to isolate the best classification method.

GitHub URL: https://github.com/DougAugSilva/IBM_Final_Project/blob/main/7_SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results



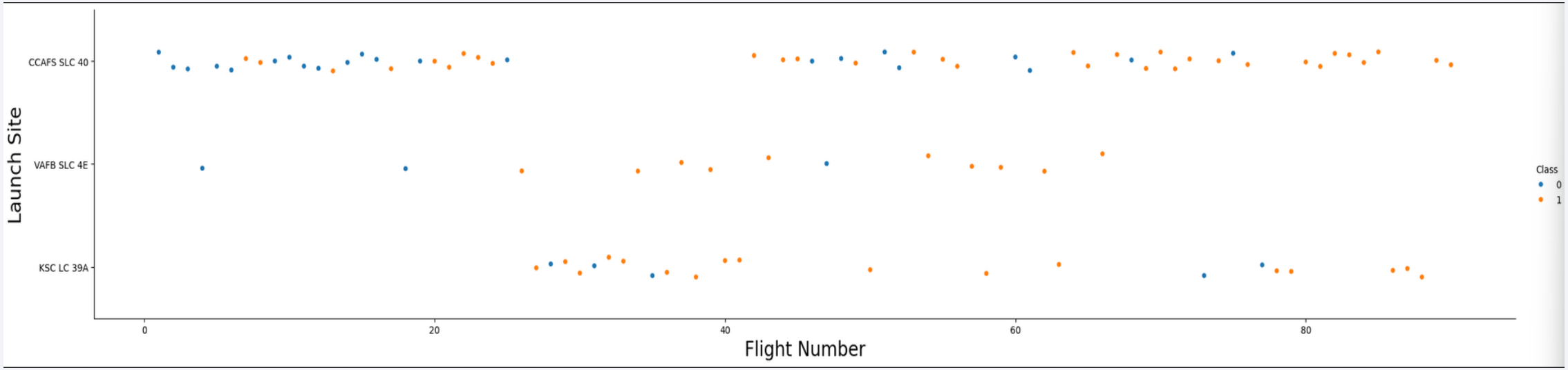


- Standardized Features: All classification models were trained using a standardized feature set comprising 83 dummy-coded parameters spanning payload mass, orbit type, launch site, and core details.
- High Training Baseline: Through 10-fold cross-validation, the Decision Tree model achieved the highest training accuracy at approximately 88.5%, while Logistic Regression, SVM, and KNN hovered around 84.6% to 84.8%.
- Equivalent Test Performance: Upon final evaluation, the models demonstrated consistent generalization with excellent predictive accuracy on the unseen 18-sample test dataset.
- Common Error Profile: The persistent challenge across models was a slight presence of false positives, meaning some failed landings were mistakenly predicted as successful.

Section 2

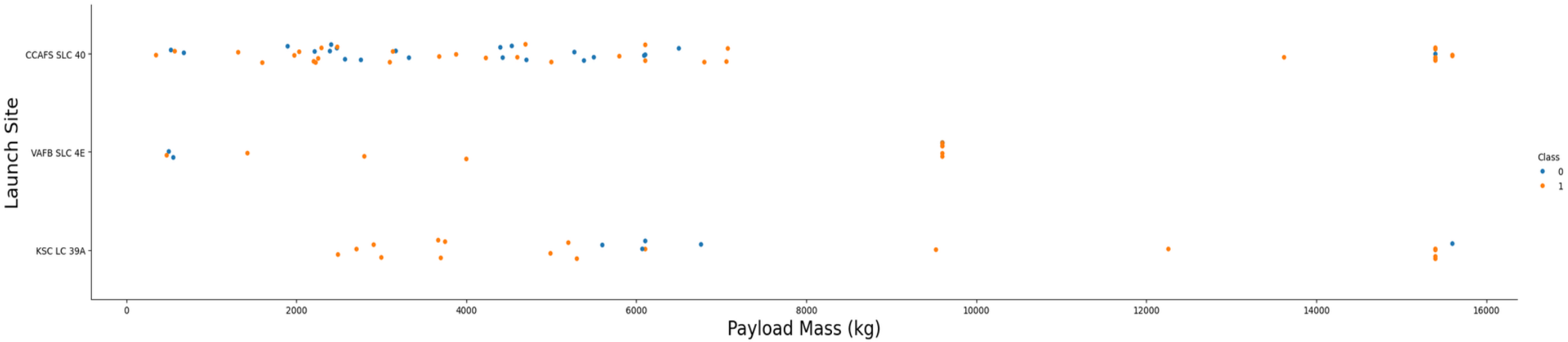
Insights drawn from EDA

Flight Number vs. Launch Site



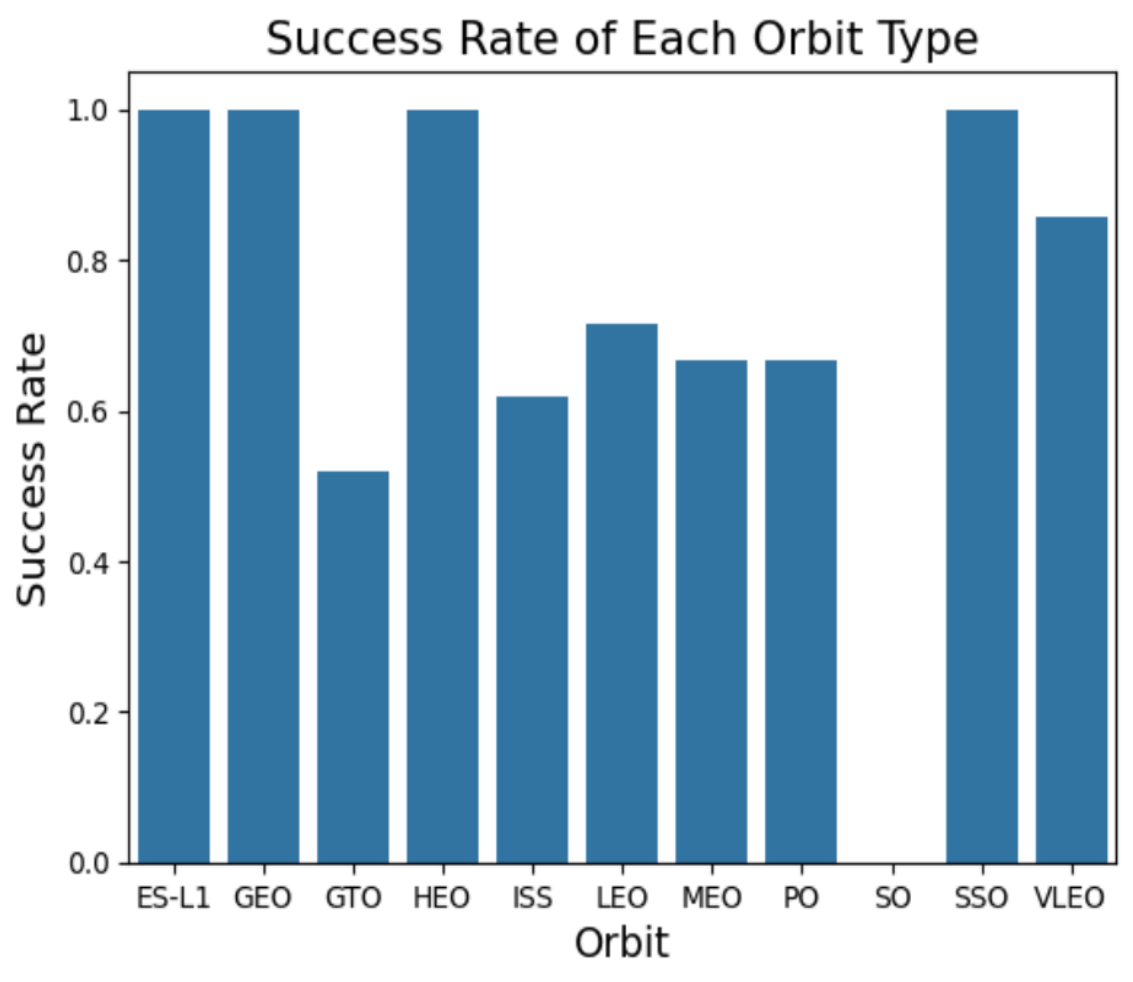
- Launch Site Utilization over Time: CCAFS SLC 40 was the primary launch site used during the earlier flights (lower flight numbers) and has been consistently used throughout the entire timeline. VAFB SLC 4E was introduced later, with launches starting around Flight Number 3. KSC LC 39A was utilized even later in the program, beginning around Flight Number 30.
- Success Rate Trend over Time: As the Flight Number increases (representing chronological progress), the density of successful landings (labeled as Class 1 / blue) increases significantly across all three launch sites. Earlier flights (mostly at CCAFS SLC 40) had a higher proportion of failures (Class 0 / orange) as SpaceX was still developing and testing its recovery technology. Recent flights across all sites show a highly dominant success rate, showcasing technological maturity and reliability.

Payload vs. Launch Site



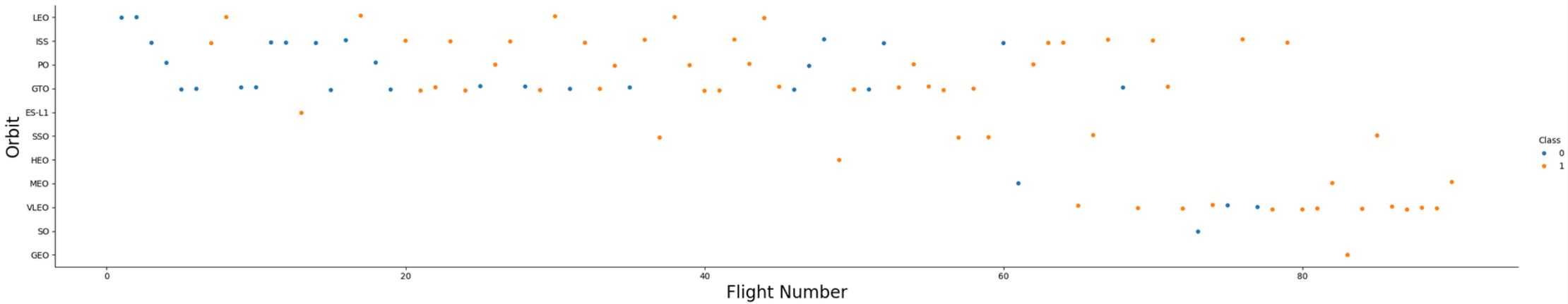
- Launch Site Payload Ranges: VAFB SLC 4E handles light to medium-heavy payloads up to approximately 9,600 kg. Notably, there are no launches with extremely heavy payloads (greater than 10,000 kg) from this site. KSC LC 39A has a clear separation: it has a cluster of lighter payloads (under 6,000 kg) and a distinct cluster of very heavy payloads (around 15,000 kg). CCAFS SLC 40 is the most versatile launch site, covering the entire range of payloads from very light (under 1,000 kg) to very heavy (up to 15,000 kg).
- Payload Mass vs. Success Rate: For heavy payloads (greater than 8,000 kg), the launch success rate (Class 1 / blue) is exceptionally high across all launch sites. For lighter payloads (less than 5,000 kg), the success rate is more mixed, especially at the older CCAFS SLC 40 site.z

Success Rate vs. Orbit Type



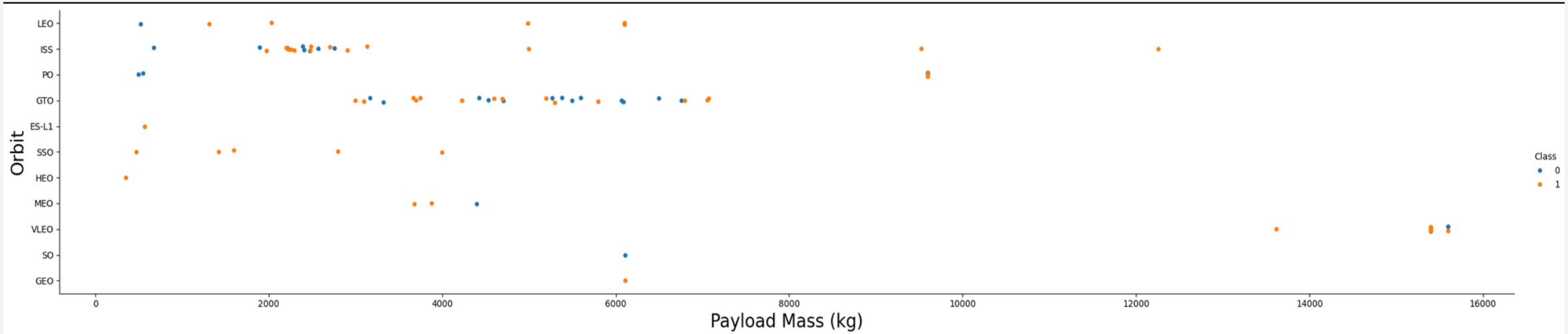
- **Perfect Success Rates (100%):** ES-L1 (Earth-Sun L1), GEO (Geostationary Orbit), HEO (Highly Elliptical Orbit), and SSO (Sun-Synchronous Orbit) all show a 100% success rate (1.0). These orbits represent highly planned scientific or critical communications missions where SpaceX achieved flawless execution.
- **High Success Rates (70% - 90%):** VLEO (Very Low Earth Orbit) and LEO (Low Earth Orbit) demonstrate strong success rates, typically hovering around 80% to 90%. MEO (Medium Earth Orbit) also shows a robust success rate of approximately 75%.
- **Moderate Success Rates (50% - 65%):** ISS (International Space Station / cargo missions), PO (Polar Orbit), and GTO (Geostationary Transfer Orbit) have success rates falling between 50% and 65%. GTO, in particular, often represents heavier payloads requiring maximum energy, making recovery slightly more challenging.
- **Lowest Success Rates:** SO (Sun Orbit) shows a 0% success rate (0.0), indicating that missions targeting this orbit in the early stage trials did not yield successful landings.

Flight Number vs. Orbit Type



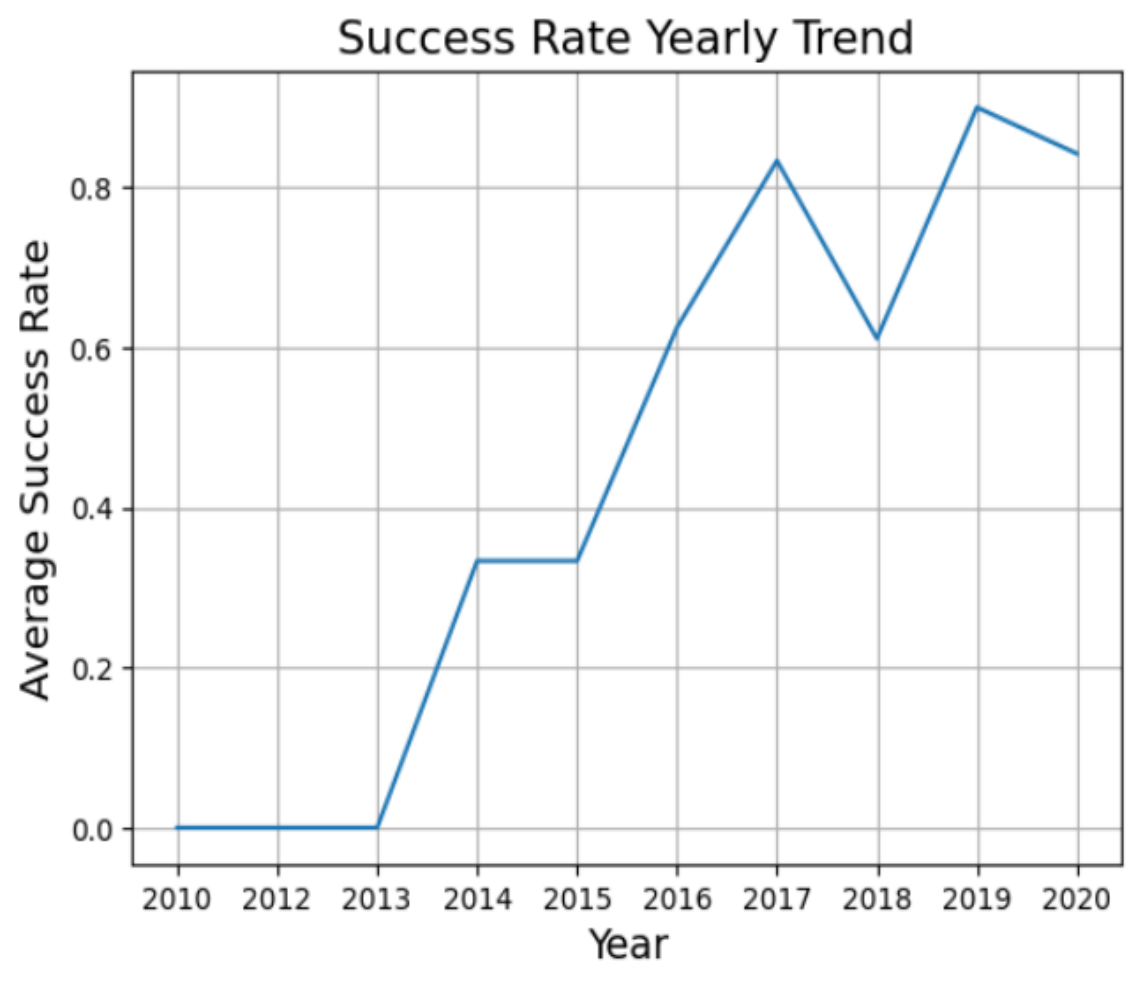
- **Orbit Sequencing and Strategy over Time:** LEO (Low Earth Orbit): This orbit was targeted frequently in the very early flight numbers (flights 1-2) but was largely paused for a long stretch until much later in the flight series (around flight 70+), when it became extremely frequent (likely due to Starlink deployments). GTO (Geostationary Transfer Orbit): This became the highly dominant target orbit in the middle period of the program (between flights 5 and 50). SSO (Sun-Synchronous Orbit), PO (Polar Orbit), and ISS: These orbits show spaced-out, periodic targeting throughout the timeline, representing specific scientific, governmental, or cargo resupply missions. VLEO (Very Low Earth Orbit): This orbit is exclusively targeted in the very late stages of the program (flight numbers 70 and beyond)
- **Success Rates (Class) over Time by Orbit: LEO:** For early LEO flights, the landing success rate was poor (mostly orange/Class 0). However, during the recent massive wave of LEO flights (high flight numbers), the landing success rate is near-perfect (Class 1). GTO: Across all flight numbers, GTO launches show a mixed set of landing outcomes (both Class 0 and Class 1 scattered throughout). This is because GTO missions are highly demanding on fuel, leaving very thin margins for a successful booster return. ISS and PO: Similar to others, earlier flights targeting these orbits had mixed success, but later flights show consistent, consecutive landing success.

Payload vs. Orbit Type



- **Payload Heavyweights vs. Lightweights:** VLEO has a highly concentrated cluster of very heavy payloads (around 15,000 kg). These are mostly SpaceX's own Starlink satellite launches, which have a near-perfect landing success rate (Class 1). GTO launches span a wide range of weights (mostly between 3,000 kg and 7,000 kg). Successful and unsuccessful landings are mixed regardless of payload mass here because GTO orbits require high velocity, leaving less margin for booster recovery. SSO, LEO, and PO primarily handle lighter payloads (under 4,000 kg). ISS payloads are mostly clustered between 2,000 kg and 4,000 kg (representing cargo missions) with solid success outcomes.
- **Payload Weight and Landing Success:** For heavy payloads (especially above 8,000 kg) destined for Polar, LEO, and ISS orbits, the landing success rate is exceptionally high. For lower payloads, success rates are more evenly distributed with failures, primarily because these include many of SpaceX's earlier experimental test flights.

Launch Success Yearly Trend



- **Early Phase (2010 - 2013):** The success rate remained at 0% (0.0) during the first few years (2010-2012). This represents the early developmental stage of the Falcon 9 rocket where experimental landing attempts were not yet successful or planned.
- **Inception of Landing Success (2013 - 2015):** In 2013, the success rate began its upward trajectory. There was a temporary plateu and slight fluctuation between 2013 and 2015 as landing methods were actively trialed in real-world scenarios.
- **Consistent Year-over-Year Growth (2015 - 2020):** Starting in 2015, there is a continuous, steep rise in the success rate. By 2020, the annual average success rate reached its peak, hovering around 85% (above 0.8).
- **Conclusion:** This dramatic rise demonstrates SpaceX's steep learning curve, showcasing how successfully recovered boosters transformed from an experimental goal into a highly reliable and standardized industry practice.

All Launch Site Names

The unique launch sites in the SpaceX dataset are:

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

SQL Query Explanation: o find these unique locations, we run a query using the DISTINCT keyword on the Launch_Site column. This ensures that even though there are many launches in the database, each launch site is listed only once in our results: SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

Launch Site Names Begin with 'CCA'

The first 5 records in the dataset where the launch site name begins with "CCA" (specifically, "CCAFS LC-40") are:

- 2010-06-04 - Booster F9 v1.0 B0003 - Payload: Dragon Spacecraft Qualification Unit
- 2010-12-08 - Booster F9 v1.0 B0004 - Payload: Dragon demo flight C1
- 2012-05-22 - Booster F9 v1.0 B0005 - Payload: Dragon demo flight C2
- 2012-10-08 - Booster F9 v1.0 B0006 - Payload: SpaceX CRS-1
- 2013-03-01 - Booster F9 v1.0 B0007 - Payload: SpaceX CRS-2

SQL Query Explanation: To retrieve these records, we use the LIKE operator with the wildcard character % to match any launch site starting with 'CCA', and limit the output to 5 rows:

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Total Payload Mass

The total payload mass carried by boosters launched by NASA (CRS) is 45,596 kg.

SQL Query Explanation: To calculate this total, we use the SUM() aggregate function on the PAYLOAD_MASS__KG_ column, filtering the dataset specifically for rows where the Customer column is exactly 'NASA (CRS)':

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 is 2,928.4 kg.

SQL Query Explanation: To compute this average, we filter the dataset using the WHERE clause for the specific booster version, and then apply the AVG() aggregate function to the PAYLOAD_MASS__KG_ column:

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass

FROM SPACEXTABLE

WHERE Booster_Version ='F9 v1.1';
```

First Successful Ground Landing Date

The date when the first successful landing outcome on a ground pad was achieved is 2015-12-22.

SQL Query Explanation: To find this date, we filter our dataset for records where the Landing_Outcome column equals 'Success (ground pad)' and then select the earliest date using the MIN() function:

```
SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing

FROM SPACEXTABLE

WHERE "Landing_Outcome" = 'Success (ground pad)';
```

Successful Drone Ship Landing with Payload between 4000 and 6000

The boosters that successfully landed on a drone ship with a payload mass greater than 4,000 kg but less than 6,000 kg are:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

SQL Query Explanation: To retrieve this list, we filter the dataset using three conditions in our WHERE clause:

- The landing outcome must be a success on a drone ship ("Landing_Outcome" = 'Success (drone ship)').
- The payload mass must be strictly greater than 4,000 kg (PAYLOAD_MASS__KG_ > 4000).
- The payload mass must be strictly less than 6,000 kg (PAYLOAD_MASS__KG_ < 6000).

```
 SELECT Booster_Version
FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Total Number of Successful and Failure Mission Outcomes

The total count of each mission outcome from the SpaceX dataset is:

- Success: 98
- Success (spacecraft-specific blank space variant): 1
- Success (payload status unclear): 1
- Failure (in flight): 1

Overall, there were 100 successful outcomes (including variants) and 1 failure outcome.

SQL Query Explanation: To calculate these counts, we use the COUNT(*) aggregate function alongside a GROUP BY clause on the Mission_Outcome column:

```
SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Boosters Carried Maximum Payload

The booster versions that have carried the absolute maximum payload mass (which is 15,600 kg) in this dataset are:

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

SQL Query Explanation: To retrieve these records, we use a subquery to first determine the highest payload mass present in the table (MAX(PAYLOAD_MASS__KG_)), and then filter our main query to find all booster versions carrying that exact mass:

```
SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ =(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

2015 Launch Records

The failed landing outcomes on a drone ship in the year 2015 are:

- January 2015 (01): Booster F9 v1.1 B1012 launched from CCAFS LC-40
- April 2015 (04): Booster F9 v1.1 B1015 launched from CCAFS LC-40

SQL Query Explanation: To retrieve these records under SQLite, we extract the year using substr(Date, 1, 4) to filter for 2015, select records matching 'Failure (drone ship)' as the Landing_Outcome, and then output the month, outcome, booster version, and launch site:

```
SELECT substr(Date, 6, 2) AS Month, "Landing_Outcome", Booster_Version, Launch_Site

FROM SPACEXTABLE

WHERE substr(Date, 1, 4) = '2015'

  AND "Landing_Outcome" = 'Failure (drone ship)';
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Between 2010-06-04 and 2017-03-20, the ranked count of landing outcomes (in descending order) is as follows:

- No attempt: 10
- Success (drone ship): 5
- Failure (drone ship): 5
- Success (ground pad): 3
- Controlled (ocean): 3
- Uncontrolled (ocean): 2
- Failure (parachute): 2
- Precluded (drone ship): 1

SQL Query Explanation: To rank these outcomes, we use the COUNT(*) aggregate function to group by the Landing_Outcome column. We apply a date filter using the BETWEEN operator to target our specific time frame and then sort the counts in descending order using ORDER BY Outcome_Count DESC:

```
SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Outcome_Count DESC;
```



Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>



In the first map generated using Folium, the following visual and interactive elements were displayed to represent the launch sites:

- **Basemap:** Rendered using Esri's map server (World Street Map), which provides a clear, accessible representation of streets, continents, oceans, and borders.
- **Orange Circles (folium.Circle):** Circles with a defined radius of 1,000 meters (1 km) are positioned around each of the four SpaceX launch sites. These visually highlight the approximate safety zone or immediate territorial extent of each location.
- **Static Text Labels (folium.map.Marker with DivIcon):** The names of the launch sites (such as CCAFS LC-40, KSC LC-39A, etc.) appear directly on the map in bold, orange text, positioned next to or over the exact coordinates of each site. This allows for immediate identification without the need to click on an icon.

<Folium Map Screenshot 2>



In the second generated map (following the completion of Task 2), the following visual and functional elements were added to the previous base map:

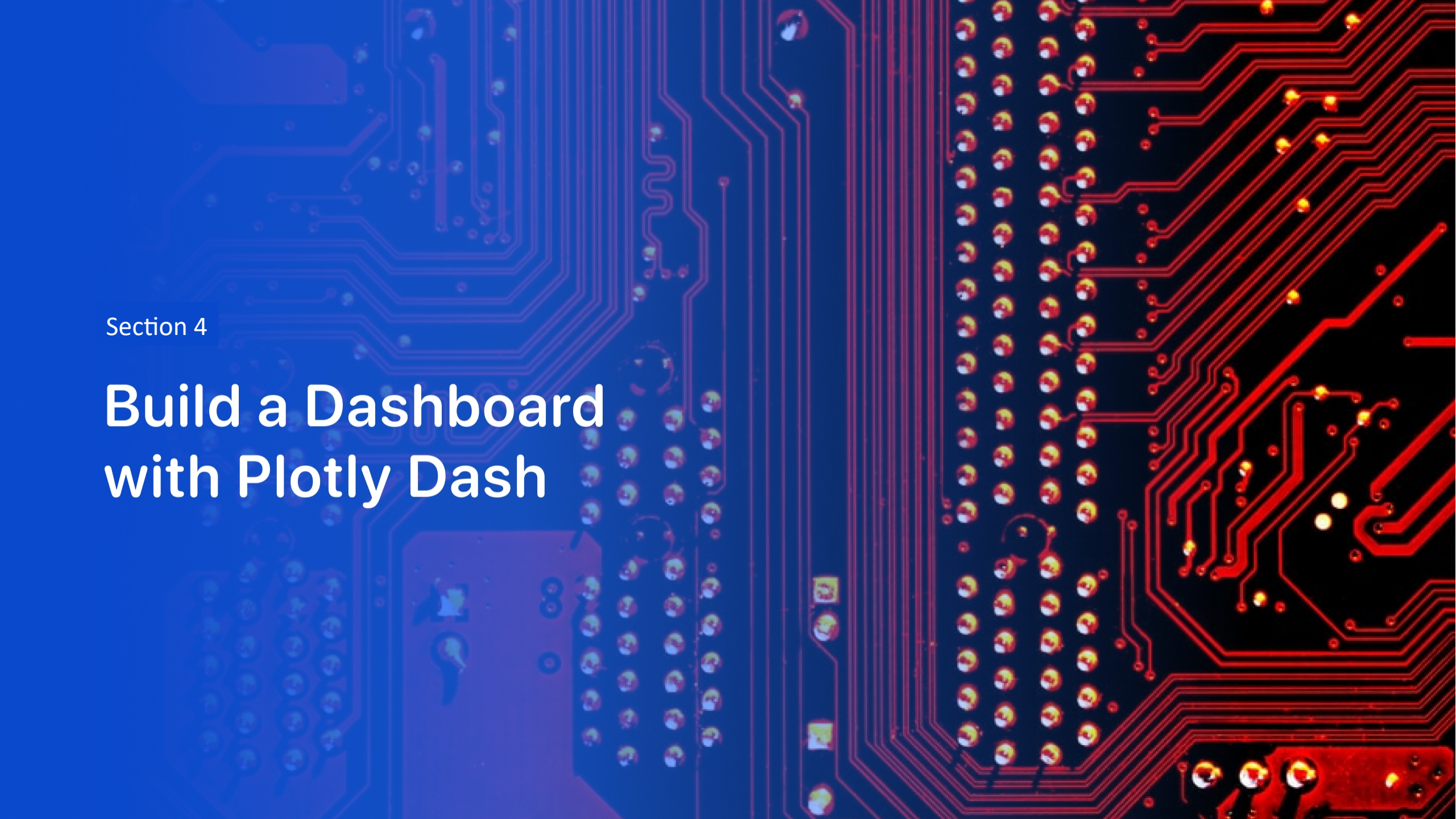
- **Marker Clusters (MarkerCluster):** Represented by colored circles containing numbers (e.g., yellow, blue, or green circles with numbers inside). These group together multiple launches that occurred at the exact same coordinates. Clicking on these clusters automatically zooms the map in to reveal the individual markers for each launch.
- **Individual Launch Markers (folium.Marker):** These are the individual location pins that appear when you zoom in on the launch sites.
- **Icon Colors (Success or Failure Indicator):** Green: Indicates that the corresponding launch was successful (class = 1). Red: Indicates that the launch failed (class = 0).

<Folium Map Screenshot 3>

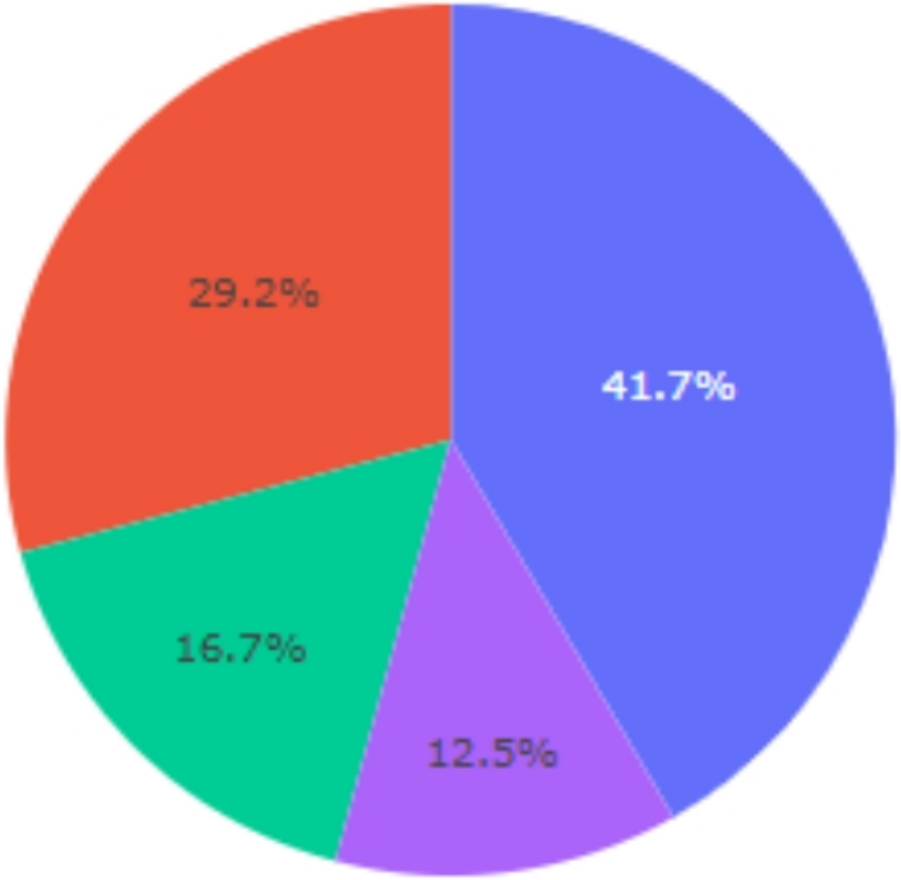


In the third map generated by Folium (focusing on Task 3), new interactive and geometric elements are introduced to analyze the distances between launch sites and their surroundings:

- **Mouse Position Plugin (MousePosition):** Displays a small box in the upper-right corner of the map, showing the exact latitude and longitude coordinates of the cursor's current position. This allows the user to easily identify the coordinates of features such as coastlines, highways, railways, and cities.
- **Custom Distance Marker (folium.Marker with DivIcon):** A marker placed over a point of interest (such as the nearest coastline) that, instead of a conventional icon, displays static text indicating the calculated distance in kilometers (e.g., 3.62 km).
- **Connection Lines (folium.PolyLine):** Straight lines drawn directly on the map, connecting the exact coordinates of a launch site to a chosen point of interest (such as a coastline, railway, or city). These lines serve as visual guides, clearly demonstrating the linear path used to calculate the geodesic distance.


Section 4
Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>



Total Succes Launches By Site

By cross-referencing the 'Class' column with the 'LaunchSite' column (as we did in the grouping and bar chart above), we can draw the following strategic conclusions:

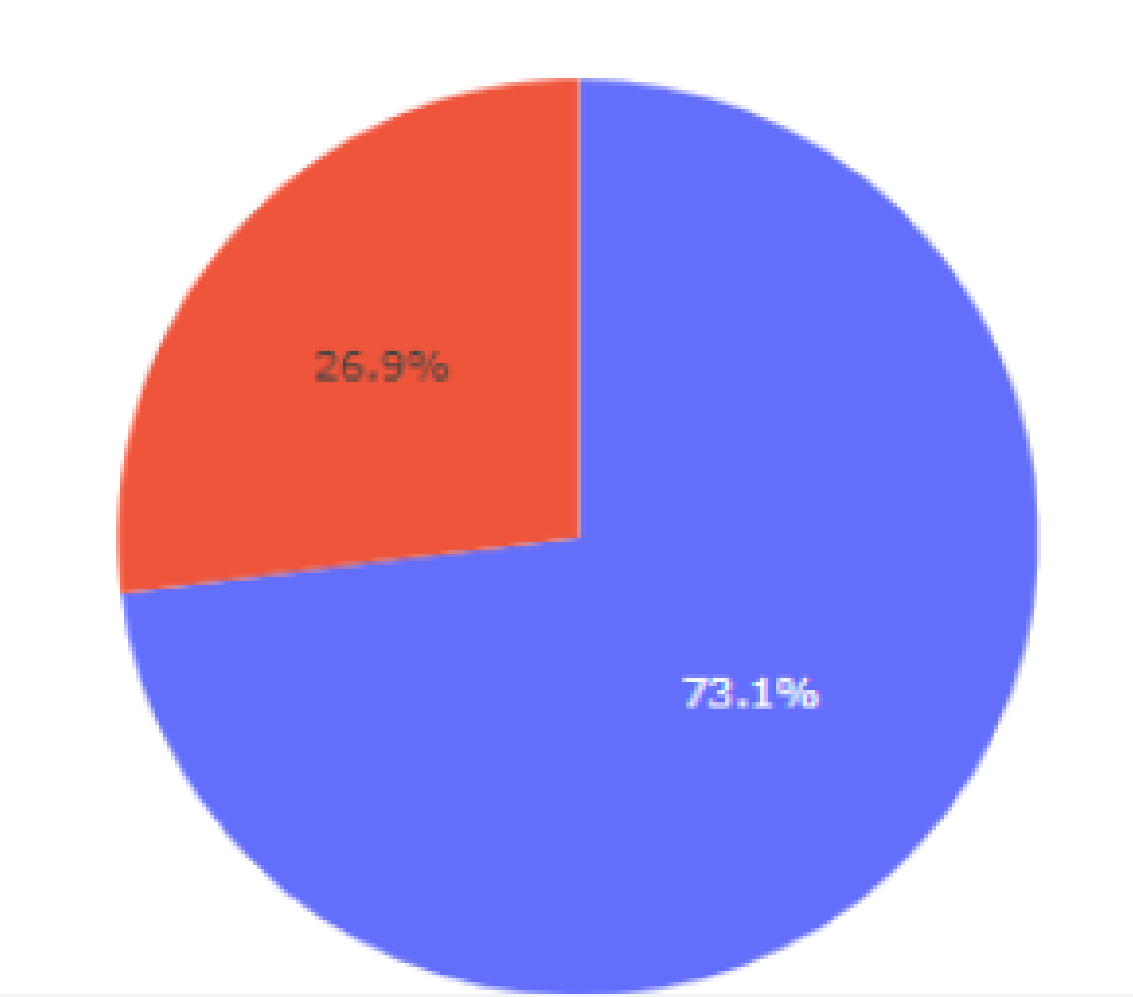
KSC LC 39A (Kennedy Space Center):

- Volume: It accounts for 22 launches.
- Performance: With 17 successes, it boasts the highest success rate in the dataset: 77.27%. This historic NASA complex began being used by SpaceX during a more mature phase of the Falcon 9 program, by which time landing technology had become more well-established.

VAFB SLC 4E (Vandenberg Space Force Base):

- Volume: This is the least-used site in the dataset, with only 13 launches (primarily focused on polar orbits).
- Performance: It records 10 successes, achieving an excellent success rate of 76.92%.

<Dashboard Screenshot 2>

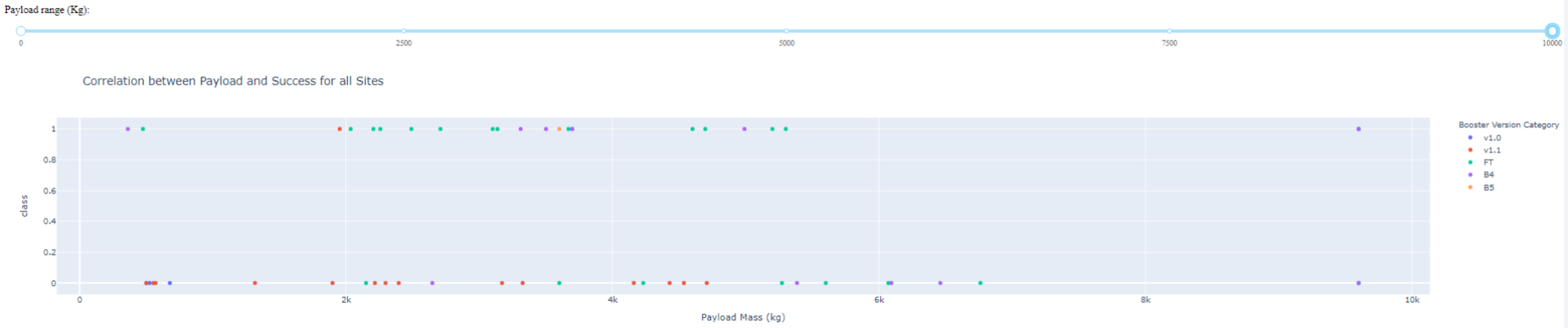


Total Succes Launches for site CCAFS LC-40

CCAFS SLC 40 (Cape Canaveral Space Force Station):

- Volume: This site has the highest number of historical launches in the dataset (55 launches).
- Performance: It records 33 successes, equating to a 60.00% success rate. Because it was heavily used during the early years of the Falcon 9 program—a period when landing failures were much more frequent—its overall historical average is somewhat lower.

<Dashboard Screenshot 3>



Correlation between Payload and Succes for all Sites

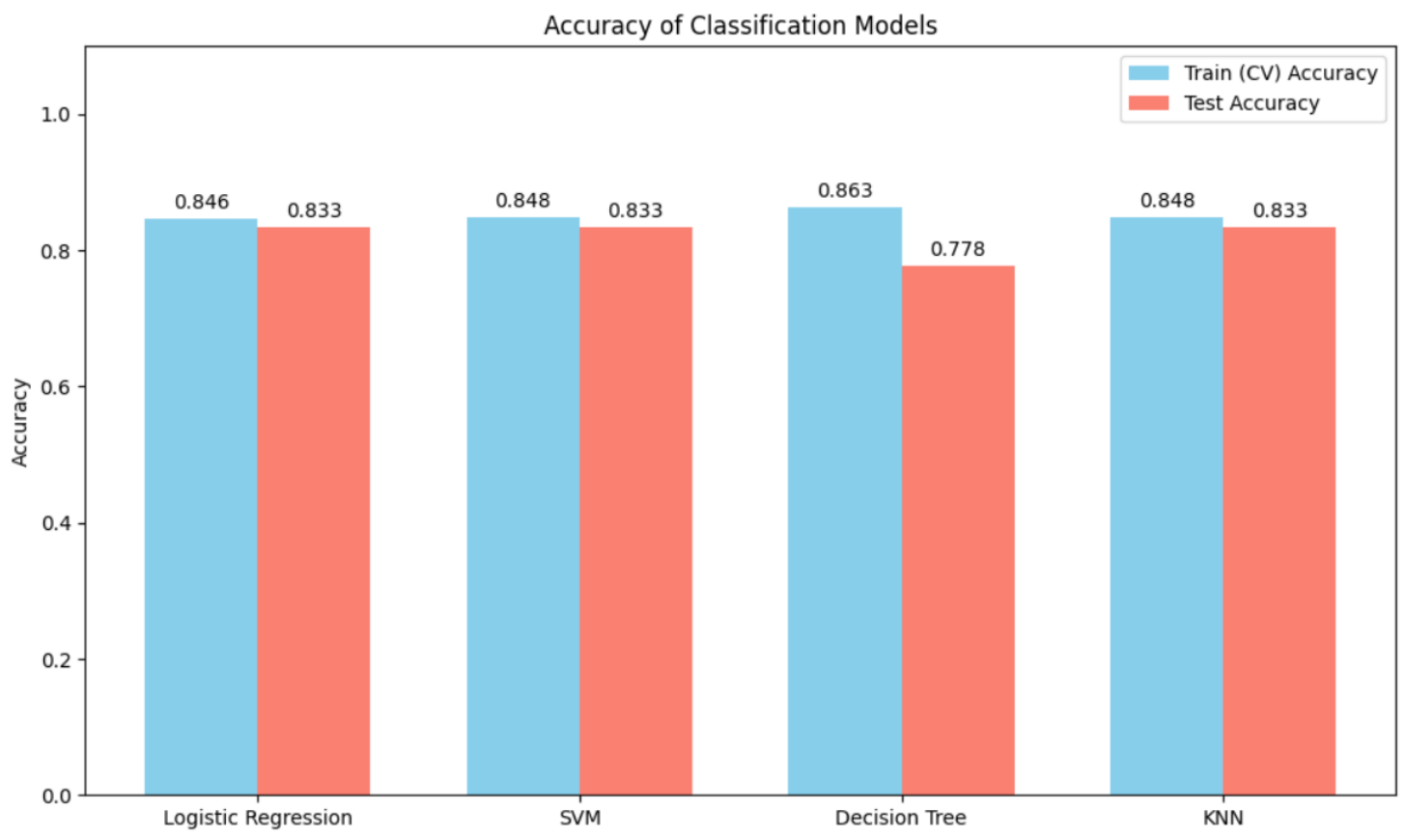
There is a strong positive correlation between heavier payload masses and landing success. However, this is heavily influenced by time and technology progression; heavier payloads were launched during the era of the highly optimized Falcon 9 Block 5 booster, which has a near-perfect recovery record.

Section 5

Predictive Analysis (Classification)



Classification Accuracy

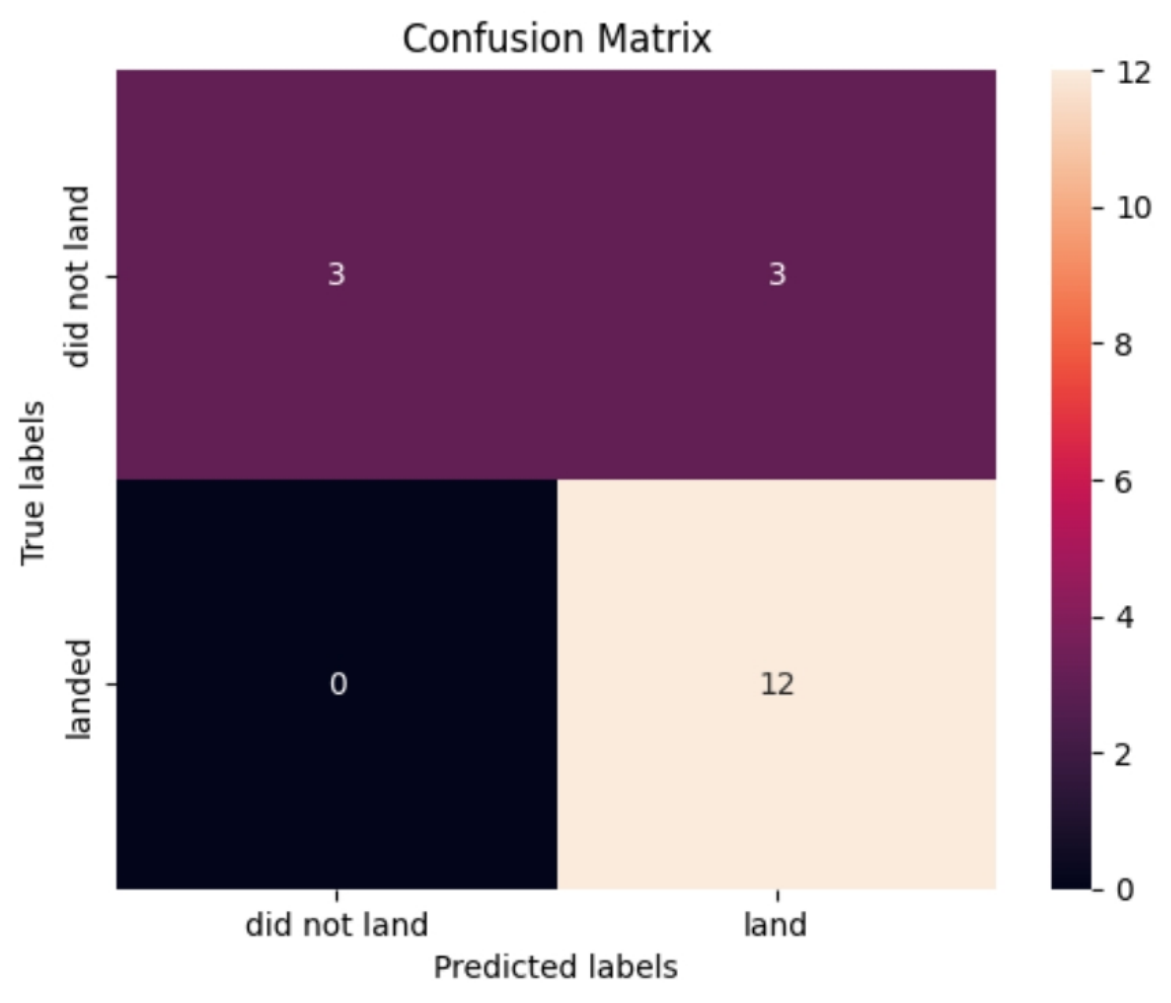


Accuracy of the models:

- Logistic Regression Test Accuracy: 0.8333
- SVM Test Accuracy: 0.8333
- Decision Tree Test Accuracy: 0.7778
- KNN Test Accuracy: 0.8333

The Logistic Regression, SVM, and KNN models tied for the highest classification accuracy on the test set, all achieving an 83.33% success rate.

Confusion Matrix



In this case, the confusion matrix belongs to the logistic regression model. The confusion matrix illustrates where the model got it right and where it got it wrong when evaluating the 18 test samples:

- True Negatives (Top-Left Corner = 3): The model correctly predicted 3 times that the rocket would not land.
- False Positives (Top-Right Corner = 3): The model incorrectly predicted 3 times that the rocket would land, when in fact it did not.
- False Negatives (Bottom-Left Corner = 0): The model made no errors of this type (it never predicted that the rocket would not land when, in fact, it did).
- True Positives (Bottom-Right Corner = 12): The model correctly predicted 12 times that the rocket would land successfully.

Conclusions

1) **Consistent Success of Classification Models:** Logistic Regression, SVM (Support Vector Machine), and KNN (K-Nearest Neighbors) algorithms all achieved an identical, robust performance on the test set, with an accuracy of 83.33%—demonstrating an excellent ability to generalize to previously unseen data.

2) **Focus Area: False Positives:** The confusion matrix revealed that the primary challenge for these classifiers lies in handling false positives (rockets that failed to land but were classified as successful landings). This "optimistic" behavior suggests that the models prioritize identifying the characteristics associated with successful landings.

3) **Fundamental Importance of Feature Engineering:** Standardizing data using `StandardScaler` and correctly encoding categorical variables (such as orbit type, launch site, and booster serial number) were crucial for enabling optimal hyperparameter tuning and preventing bias stemming from differences in scales, such as mass and flight count.

4) **Geographic and Operational Impact:** Historical analysis shows that the success rate varies significantly depending on the launch site (e.g., Kennedy Space Center's LC-39A versus SLC-40) and that lower-altitude orbits (LEO, VLEO, ISS) historically yield better statistics for stage recovery.

5) **Technological Evolution and Reusability:** The landing success rate is directly correlated with technological maturity and the booster's version (Block). Newer rockets—as well as those that have already undergone reuse—demonstrate a significantly higher probability of successful landings compared to early prototypes.

6) **Economic Viability for Decision-Making:** With the forecasting precision achieved through machine learning models, SpaceX's competitors or clients can determine—with a high degree of reliability—the approximate operational cost of a launch ($62 million for reused flights, compared to over $165 million for competitors' expendable rockets), serving as a powerful strategic tool for commercial proposals.

Appendix

List of all Jupyter notebooks and Markdown files used in this presentation:

- **Notebook 1:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/1_jupyter-labs-spacex-data-collection-api.ipynb
- **Notebook 2:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/2_jupyter-labs-webscraping.ipynb
- **Notebook 3:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/3_labs-jupyter-spacex-Data%20wrangling.ipynb
- **Notebook 4:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/4_jupyter-labs-eda-sql-coursera_sqllite.ipynb
- **Notebook 5:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/5_eda_data_viz.ipynb
- **Notebook 6:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/6_lab_jupyter_launch_site_location.ipynb
- **Notebook 7:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/7_SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb
- **Markdown:** https://github.com/DougAugSilva/IBM_Final_Project/blob/main/Dashboard_Instructions.md

Thank you!

